

Department of Mathematics, IIT Madras
MA-5105-Scientific Computing & Advanced Programming Lab
Practice Problems 1

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Q. 1 Initialise the string "Barking up the wrong tree", preferably with some variable name of your choice

- access the element number 15
- concatenate it with "in futility"
- display the concatenated string
- print its length using the `len()` function.
- what happens when you try to access the element no 54.
- eventually consolidate all the above in a single script and run.

Q. 2 try negative indices lets say `[-9:-6]` for a string.

Q. 3 Write a script that assigns the floating-point literal 175000.0 to the variable `num` using exponential notation, and then prints `num` in the interactive window.

Q. 4 Now use concatenation to create a new string from an old string by changing only a few letters.

Q. 5 Write a script called `temperature.py` that defines two functions:

- `convert_cel_to_far()` which takes one float parameter representing degrees Celsius and returns a float representing the same temperature in degrees Fahrenheit using the following formula:

$$F = C * 9/5 + 32$$

- `convert_far_to_cel()` which take one float parameter representing degrees Fahrenheit and returns a float representing the same temperature in degrees Celsius using the following formula:

$$C = (F - 32) * 5/9$$

- The script should first prompt the user to enter a temperature in degrees Fahrenheit and then display the temperature converted to Celsius.

- Then prompt the user to enter a temperature in degrees Celsius and display the temperature converted to Fahrenheit.
- All converted temperatures should be rounded to 2 decimal places.